25901918 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY201 - ESSENTIAL MATHEMATICS FOR PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **3** marks

1. State the physical significance of the scalar (dot) product of two vectors. Give one practical example. [U] / [CO1]

2. Explain the **geometrical meaning** of the inner product in Euclidean space. [K] / [CO1, CO7]

3. When is the scalar triple product zero? What is the geometrical implcation? [U] / [CO1, CO7]

4. $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ Find the eigen vectors of matrix **A.** [A] / [CO3, CO7]

5. Explain why all diagonal elements of a Hermitian matrix are real. [A] / [CO2, CO7]

6. What is meant by a similarity transformation? Explain. [U] / [CO3]

7. In Cartesian coordinates system the Scale Factors $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$. Write down the equation for the Scale Factors in General curvilinear coordinate system. [U] / [CO5]

8. The displacement vector of a particle is given by $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$. Write the expression for the time derivative of this vector and explain its physical meaning. [U] / [CO4, CO7]

9. Prove that $\nabla \cdot (\nabla \times \vec{A}) = 0$ [A] / [CO4]

10. A conservative force field has potential function $\varphi(x, y) = x^3y - xy^3 + 2y$. Find the force field F and calculate the work done in moving from $(1,1) to (2,-1)$ [A] / [CO6, CO7]

11. What the divergence theorem states about flux and volume. [U] / [CO6]

12. Write the expression for the volume integral of a scalar function $f(x, y, z)$ over a volume V [U] / [CO6]

[3x10 = 30]

Part B

Short Essay Type Questions

Answer any **8** questions

Each question carries **5** marks

13. Write the statement of the triangle inequality in vector spaces and check its validity for the vectors (3,−1,2) and (−1,4,0) [A] / [CO1, CO7]

14. Distinguish between orthogonal and orthonormal vectors. Orthogonalize the vectors (0, 1, 0) and (1, 0, 1). [A] / [CO1, CO7]

15. Solve the equations using matrices: [A] / [CO2]

$$x - y = 1,\ x + y = 5$$

16. Show that $\boldsymbol{\sigma}_x\,\boldsymbol{\sigma}_y - \boldsymbol{\sigma}_y\,\boldsymbol{\sigma}_y = 2i\,\boldsymbol{\sigma}_z$ Where $\boldsymbol{\sigma}_x$, $\boldsymbol{\sigma}_y$ and $\boldsymbol{\sigma}_z$ be the Pauli spin matrices. $\sigma_{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_{\mathbf{y}} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_{\mathbf{z}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$ [A] / [CO2, CO7]

17. $\mathbf{A} = \frac{1}{3}\begin{bmatrix} 2-i & 2 \\ 2 & 2+i \end{bmatrix}$ Determine whether the given matrix is unitary, also calculate its determinant. [U] / [CO3, CO7]

18. The gradient of a scalar field in spherical polar coordinates is given by $\nabla\phi = \frac{\partial\phi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\phi}\hat{\phi}$ Find $\nabla\phi$ if $\phi(r, \theta, \phi) = r^2 \sin\theta\cos\phi$ [A] / [CO5]

19. $\phi = x^2y$ and $\psi = 2yz^3$. Verify the identity $\nabla(\phi\psi) = \psi(\nabla\phi) + \phi(\nabla\psi)$. [A] / [CO4, CO7]

20. A vector field is $\vec{A} = yz\,\hat{i} + xz\,\hat{j} + xy\,\hat{k}$. Find the divergence of the vector field and explain its physical meaning. [U] / [CO4, CO7]

21. Derive the formula for the total surface area of a sphere of radius r using spherical coordinates. [A] / [CO6]

22. A particle moves along a straight line from $x = 0 \quad \text{to} \quad x = 2$ under the influence of a force given by $F(x) = 3x + 4$ (in newtons). Calculate the work done by the force over this distance. [A] / [CO6]

[5x8 = 40]